



MLC School

2016

YEAR 12 TRIAL HSC EXAMINATION

Mathematics Extension 1

Name:

Teacher:

Date: 2016

Weighting: 40 %

General Instructions

- Reading time – 5 minutes
- Working time – 2 hours
- Write in blue or black pen
- Board approved calculators may be used
- All necessary working should be shown in questions 11 - 14
- A separate reference sheet is provided

Total Marks – 70

Section 1 Pages 3 - 5

10 marks

Attempt Questions 1 - 10

Allow about 15 minutes for this section

Section 2 Pages 6 - 10

60 marks

Attempt Questions 11 - 14

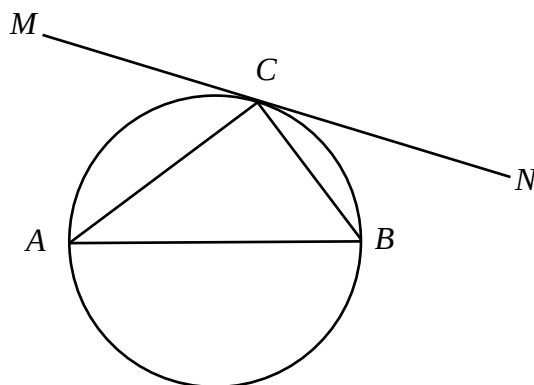
Allow about 1 hour 45 minutes for this section

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Section I**10 marks****Attempt Questions 1-10****Allow 15 minutes for this section**

Use the multiple-choice answer sheet for Questions 1-10

- 1** In the diagram, AB is a diameter of the circle and MCN is the tangent to the circle at C. $\angle CAB = 35^\circ$. What is the size of $\angle MCA$?



- (A) 35°
(B) 45°
(C) 55°
(D) 65°
- 2** Which of the following is the domain of $y = \cos^{-1}\left(\frac{x}{2}\right)$?
- (A) $-2 \leq x \leq 2$
(B) $0 < x < 2\pi$
(C) $-\frac{1}{2} \leq x \leq \frac{1}{2}$
(D) $0 \leq x \leq \frac{\pi}{2}$

- 3 The solution to the inequality $\frac{x-1}{x+2} > 0$ is:
- (A) $-2 < x < 1$
(B) $-1 < x < 2$
(C) $x < -2$ or $x > 1$
(D) $x < -1$ or $x > 2$
- 4 The acute angle between the lines $2x - y = 0$ and $kx - y = 0$ is equal to $\frac{\pi}{4}$.
What is the value of k ?
- (A) $k = -3$ or $k = -\frac{1}{3}$
(B) $k = -3$ or $k = \frac{1}{3}$
(C) $k = 3$ or $k = -\frac{1}{3}$
(D) $k = 3$ or $k = \frac{1}{3}$
- 5 After t years the number N of individuals in a population is given by $N = 400 + 100e^{-0.1t}$.
What is the difference between the initial population size and the limiting population size?
- (A) 100
(B) 300
(C) 400
(D) 500
- 6 The point dividing the interval from $A(-3,1)$ to $B(1,-1)$ externally in the ratio 3:1 is:
- (A) $(0, -\frac{1}{2})$
(B) $(-1, \frac{1}{2})$
(C) $(-5, 2)$
(D) $(3, -2)$

- 7 The expression $\frac{\sin 3\theta}{\sin \theta} - \frac{\cos 3\theta}{\cos \theta}$ can be simplified to
- (A) $\sin 2\theta - \cos 2\theta$
(B) $\sin 2\theta + \cos 2\theta$
(C) $\tan 2\theta$
(D) 2
- 8 Which of the following is an asymptote of the curve $y = \frac{x^2 - 4}{x}$?
- (A) $y = x$
(B) $x = 2$
(C) $x = 1$
(D) $y = 0$
- 9 Which of the following is an expression for $1 + \sec x$ in terms of t given $t = \tan \frac{x}{2}$?
- (A) $\frac{2}{1+t^2}$
(B) $\frac{2}{1-t^2}$
(C) $\frac{2t}{1+t^2}$
(D) $\frac{2t}{1-t^2}$
- 10 Which of the following is a solution of the equation $2^x = 5$?
- (A) $x = \sqrt{5}$
(B) $x = \log_e x$
(C) $x = \frac{\log_e 5}{\log_e 2}$
(D) $x = \frac{\log_e 2}{\log_e 5}$

Section II**60 marks****Attempt Questions 11-14****Allow 1 hour 45 minutes for this section**

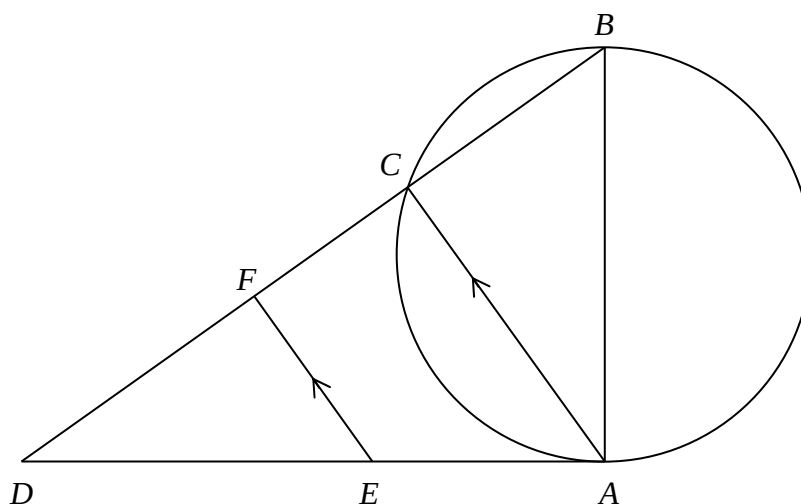
Answer in separate writing booklets for this section. Start each question in a new booklet.

Question 11. (15 marks)**Marks**

(a) Find $\lim_{x \rightarrow 0} \frac{\sin 3x}{x}$.

2

(b)



AB is a diameter of the circle and C is a point on the circle. The tangent to the circle at A meets BC produced at D . E is a point on AD and F is a point on CD such that EF is parallel to AC .

- | | | |
|-------|--|----------|
| (i) | Give a reason why $\angle EAC = \angle ABC$. | 1 |
| (ii) | Hence or otherwise show that $EABF$ is a cyclic quadrilateral. | 2 |
| (iii) | Explain why BE is a diameter of the circle through E, A, B and F . | 1 |

(Question 11 continues on the next page)

(Question 11 continued)

Marks

(c) Solve for x : $\frac{2x-3}{x} \leq 4$ **2**

(d) Use Mathematical induction to show that for all positive integers $n \geq 2$,

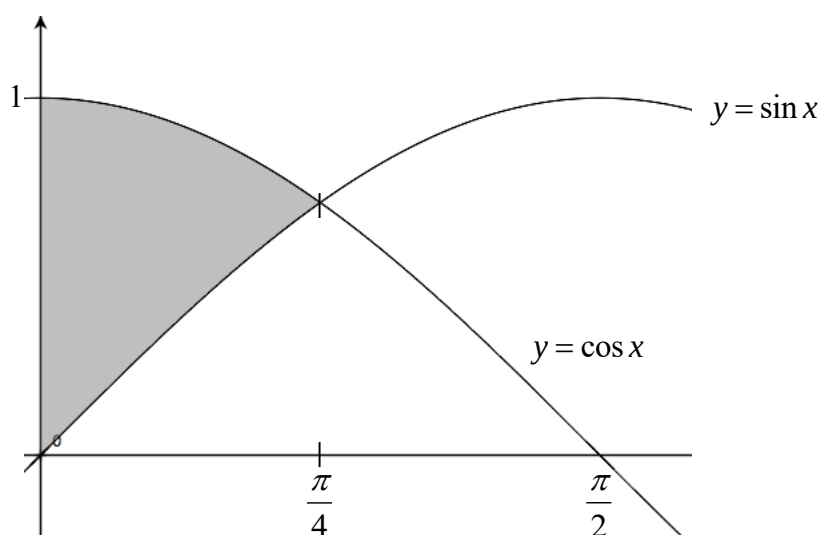
$$2 \times 1 + 3 \times 2 + 4 \times 3 + \dots + n(n-1) = \frac{n(n^2-1)}{3}$$
4

(e) Find the exact value of $\int_{\sqrt{2}}^{\sqrt{3}} \frac{1}{\sqrt{4-x^2}} dx$ **2**

Question 12. (15 marks) (Start a new booklet)

Marks

(a)



The region bounded by the curves $y = \sin x$ and $y = \cos x$ between $x = 0$ and $x = \frac{\pi}{4}$ is rotated through one complete revolution around the x -axis. Find the volume of the solid of revolution.

3

(b) (i) Show that $\frac{d}{dx}(\sin^2 x) = \sin 2x$ **1**

(ii) Hence use the substitution $u = \sin^2 x$ to evaluate $\int_{\pi/4}^{\pi/3} \frac{\sin 2x}{1 + \sin^2 x} dx$ **3**

(Question 12 continues on the next page)

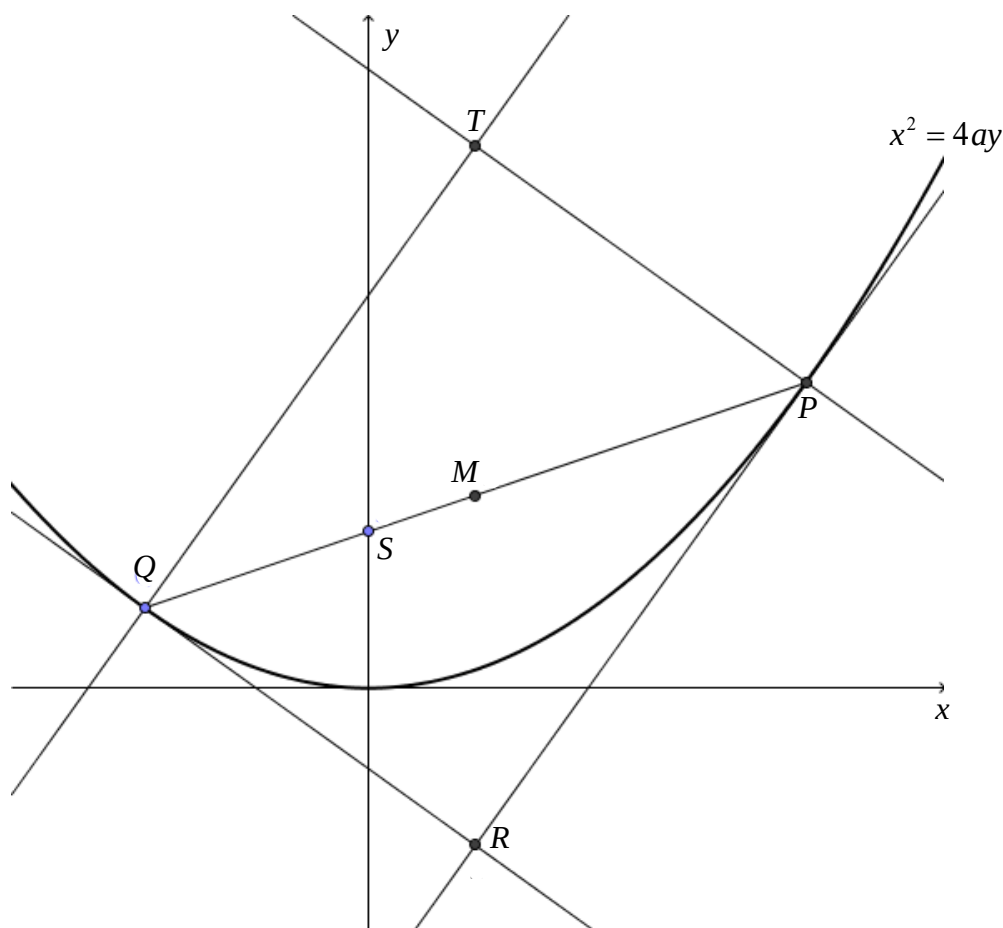
(Question 12 continued)

Marks

(c) Find $\int \frac{1+2x}{1+x^2} dx$.

3

(d)



The points $P(2ap, ap^2)$ and $Q(2aq, aq^2)$ lie on the parabola $x^2 = 4ay$. The tangents to the parabola at P and Q intersect at R . The normals at P and Q intersect at T . The point M is the midpoint of the chord PQ . The point S is the focus $(0, a)$.

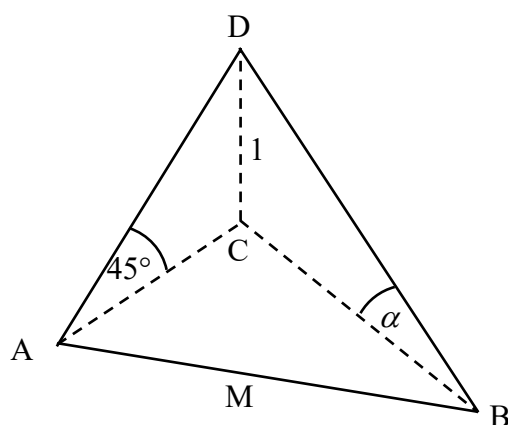
- (i) Find the coordinates of M . **1**
- (ii) Show that $pq = -1$ if PQ is a focal chord. **2**
- (iii) By considering the x -coordinates of the three points or otherwise, show that if PQ is a focal chord, then the points R , T and M are collinear. **3**

Question 13. (15 marks) (Start a new booklet)**Marks**

- (a) Consider the function $f(x) = (x+2)^2 - 9$, $-2 \leq x \leq 2$.
- (i) Find the equation of the inverse function $f^{-1}(x)$. **1**
- (ii) On the same diagram, sketch the graphs of $y = f(x)$ and $y = f^{-1}(x)$, showing clearly the coordinates of the endpoints and the intercepts on the coordinate axes. **3**
- (iii) Find the x -coordinate of the point of intersection of the curves $y = f(x)$ and $y = f^{-1}(x)$. **2**

- (b) Consider the function $f(x) = \tan^{-1}(x-1)$.
- (i) Sketch the curve $y = f(x)$, showing clearly the equations of any asymptotes and the intercepts on the coordinate axes. **2**
- (ii) Find the equation of the tangent to the curve $y = f(x)$ at the point where $x = 1$. **2**

(c)



CD is a vertical pole of height 1 metre that stands with its base C on horizontal ground. A is a point due South of C such that the angle of elevation of D from A is 45° . B is a point due East of C such that the angle of elevation of D from B is α . M is the midpoint of AB.

- (i) Show that $BC = \cot \alpha$ and hence show that $AB = \operatorname{cosec} \alpha$. **3**
- (ii) Show that $CM = \frac{1}{2} \operatorname{cosec} \alpha$ **2**

Question 14. (15 marks) (Start a new booklet) Marks

- (a) A particle is performing Simple Harmonic Motion in a straight line. At time t seconds, it has displacement x metres from a fixed point O in the line, velocity $v \text{ ms}^{-1}$ given by $v = -12 \sin\left(2t + \frac{\pi}{3}\right)$ and acceleration $\ddot{x} \text{ ms}^{-2}$. Initially the particle is 5 metres to the right of O .
- (i) Find an expression for x . 1
 - (ii) Show that $\ddot{x} = -4(x - 2)$. 2
 - (ii) Find the extremes of motion. 2
 - (iii) Find the time taken by the particle to return to its starting point for the first time. 2
- (b) After t hours, the number of individuals in a population is given by $N = 500 - 400e^{-0.1t}$.
- (i) Sketch the graph of N as a function of t , showing clearly the initial population size and the limiting population size. 2
 - (ii) Show that $\frac{dN}{dt} = 0.1(500 - N)$. 1
 - (iii) Find the population size for which the rate of growth of the population is half the initial rate of growth. 1
- (c) A particle is moving in a straight line. After time t seconds, it has displacement x metres from a fixed point O in the line, velocity $v \text{ ms}^{-1}$ given by $v = \sqrt{x}$, and acceleration $a \text{ ms}^{-2}$. Initially the particle is 1 metre to the right of O .
- (i) Show that a is constant. 1
 - (ii) Express x in terms of t . 2
 - (iii) Find the distance travelled by the particle during the third second of motion. 1

END OF EXAM



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SOLUTIONS

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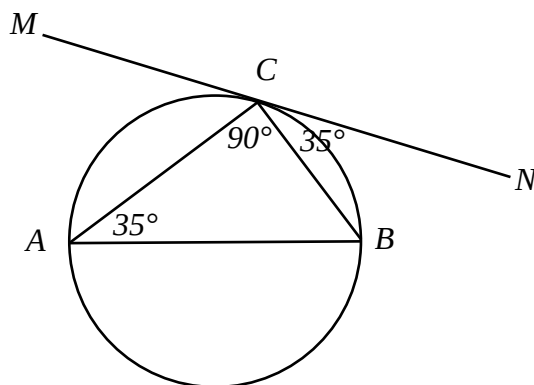
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What is the value of k ?

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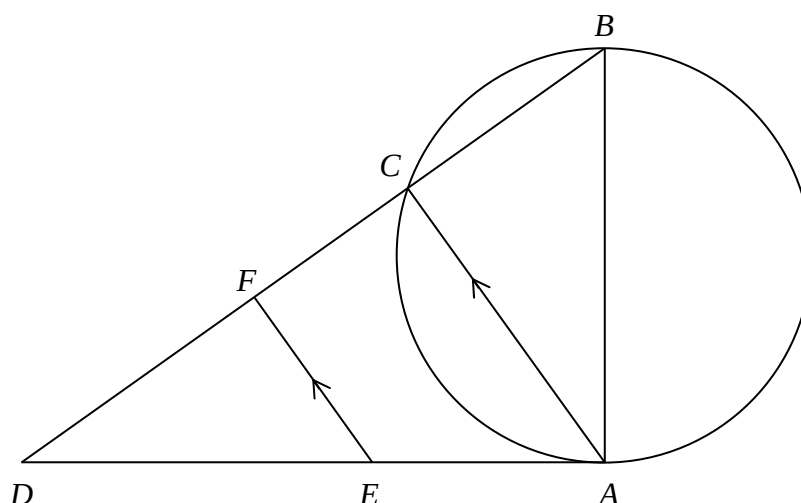
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(a) Find $\lim_{x \rightarrow 0} \frac{\sin 3x}{x}$.

2

$$\lim_{x \rightarrow 0} \frac{\sin 3x}{x} = 3 \lim_{x \rightarrow 0} \frac{\sin 3x}{3x} = 3$$

(b)



AB is a diameter of the circle and C is a point on the circle. The tangent to the circle at A meets BC produced at D . E is a point on AD and F is a point on CD such that EF is parallel to AC .

- (i) Give a reason why $\angle EAC = \angle ABC$.

1

Angle between tangent and chord equals angle in alternate segment

- (ii) Hence or otherwise show that $EABF$ is a cyclic quadrilateral.

2

$$\angle DEF = \angle EAC \quad (\text{corresponding angles in } \parallel \text{ lines})$$

$$\angle DEF = \angle ABC$$

$\therefore EABF$ is cyclic (exterior angle equals opposite interior angle)

Marks

- (iii) Explain why
- BE
- is a diameter of the circle through
- E
- ,
- A
- ,
- B
- and
- F
- .

1

$$\angle EAB = 90^\circ \quad (\text{tangent perpendicular to radius})$$

$\therefore BE$ is a diameter of circle $EABF$ (angle in semi-circle = 90°)

- (c) Solve for
- x
- :
- $\frac{2x-3}{x} \leq 4$

2

$$x(2x-3) \leq 4x^2$$

$$x(2x-3) - 4x^2 \leq 0$$

$$x(2x-3-4x) \leq 0$$

$$-x(3+2x) \leq 0$$

$$x > 0 \quad \text{or} \quad x \leq -\frac{3}{2}$$

- (d) Use Mathematical induction to show that for all positive integers
- $n \geq 2$
- ,

$$2 \times 1 + 3 \times 2 + 4 \times 3 + \dots + n(n-1) = \frac{n(n^2-1)}{3}$$

4

Let $n = 2$: $2 \times 1 = \frac{2(2^2-1)}{3} = 2 \quad \therefore \text{true when } n = 2$

Assume true for $n = k$: $2 \times 1 + 3 \times 2 + 4 \times 3 + \dots + k(k-1) = \frac{k(k^2-1)}{3}$

Let $n = k + 1$ and show that

$$2 \times 1 + 3 \times 2 + 4 \times 3 + \dots + k(k-1) + (k+1)k = \frac{(k+1)((k+1)^2-1)}{3}$$

$$LHS = 2 \times 1 + 3 \times 2 + 4 \times 3 + \dots + k(k-1) + (k+1)k$$

$$= \frac{k(k^2-1)}{3} + k(k+1)$$

$$= \frac{k((k^2-1) + 3(k+1))}{3}$$

$$= \frac{k(k^2 + 3k + 2)}{3}$$

$$= \frac{k(k+2)(k+1)}{3}$$

$$= RHS$$

$\therefore$ true for $n = k + 1$ if true for $n = k$

$\therefore$ true for all positive integers $n \geq 2$

(e) Find the exact value of $\int_{\sqrt{2}}^{\sqrt{3}} \frac{1}{\sqrt{4-x^2}} dx$

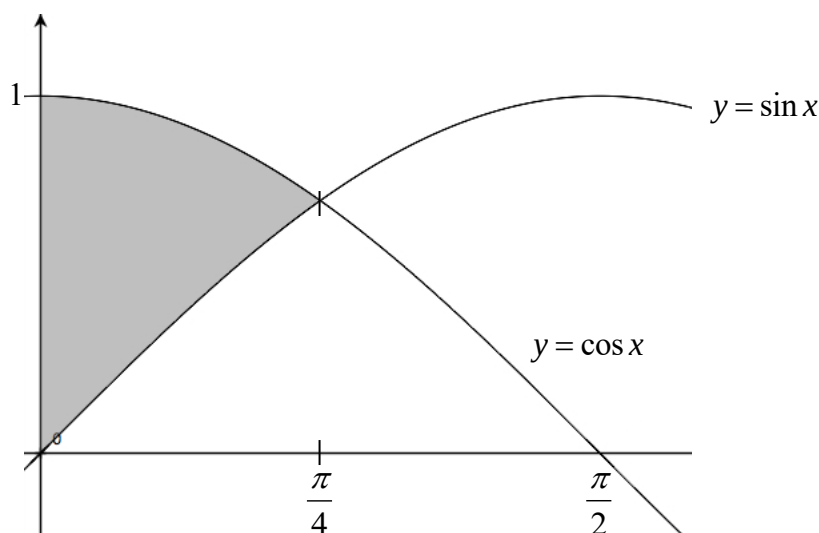
2

$$\begin{aligned} \int_{\sqrt{2}}^{\sqrt{3}} \frac{1}{\sqrt{4-x^2}} dx &= \left[\sin^{-1} \frac{x}{2} \right]_{\sqrt{2}}^{\sqrt{3}} \\ &= \left(\sin^{-1} \frac{\sqrt{3}}{2} \right) - \left(\sin^{-1} \frac{\sqrt{2}}{2} \right) \\ &= \frac{\pi}{3} - \frac{\pi}{4} \\ &= \frac{\pi}{12} \end{aligned}$$

Question 12. (15 marks) (Start a new booklet)

Marks

(a)



The region bounded by the curves $y = \sin x$ and $y = \cos x$ between $x = 0$ and $x = \frac{\pi}{4}$ is rotated through one complete revolution around the x -axis. Find the volume of the solid of revolution.

3

$$\begin{aligned} V &= \pi \int_0^{\pi/4} (\cos^2 x - \sin^2 x) dx \\ &= \pi \int_0^{\pi/4} \cos 2x dx \\ &= \frac{\pi}{2} \left[\sin 2x \right]_0^{\pi/4} \\ &= \frac{\pi}{2} \left(\sin \frac{\pi}{2} - \sin 0 \right) \\ &= \frac{\pi}{2} \end{aligned}$$

(b) (i) Show that $\frac{d}{dx}(\sin^2 x) = \sin 2x$ 1

$$\frac{d}{dx}(\sin^2 x) = 2 \sin x \cos x = \sin 2x$$

(ii) Hence use the substitution $u = \sin^2 x$ to evaluate $\int_{\pi/4}^{\pi/3} \frac{\sin 2x}{1 + \sin^2 x} dx$ 3

$$u = \sin^2 x \Rightarrow du = 2 \sin x \cos x dx = \sin 2x dx$$

$$x = \frac{\pi}{3} \Rightarrow u = \frac{3}{4}$$

$$x = \frac{\pi}{4} \Rightarrow u = \frac{1}{2}$$

$$\int_{\pi/4}^{\pi/3} \frac{\sin 2x}{1 + \sin^2 x} dx = \int_{1/2}^{3/4} \frac{1}{1 + u} du$$

$$= \left[\ln(1 + u) \right]_{1/2}^{3/4}$$

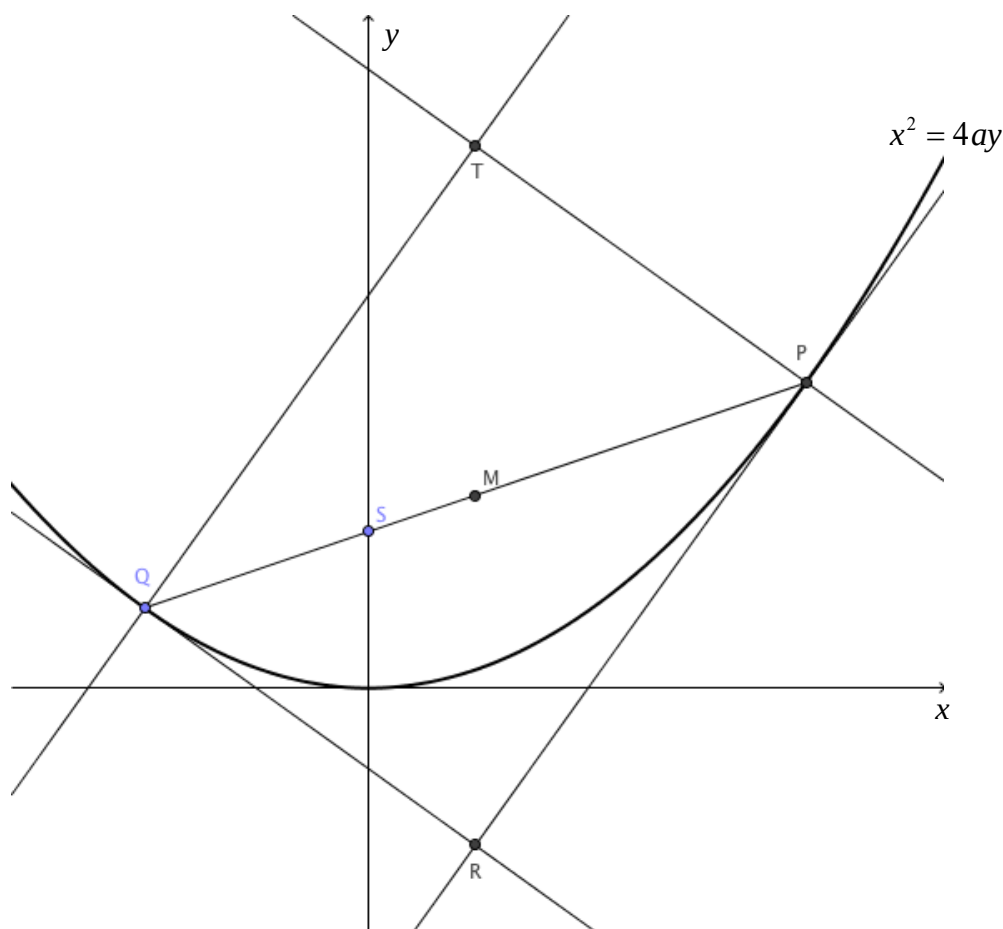
$$= \ln \frac{7}{4} - \ln \frac{3}{2}$$

$$= \ln \frac{7}{6}$$

(c) Find $\int \frac{1 + 2x}{1 + x^2} dx$. 3

$$\begin{aligned} \int \frac{1 + 2x}{1 + x^2} dx &= \int \frac{1}{1 + x^2} dx + \int \frac{2x}{1 + x^2} dx \\ &= \tan^{-1} x + \ln(1 + x^2) + c \end{aligned}$$

(d)



The points $P(2ap, ap^2)$ and $Q(2aq, aq^2)$ lie on the parabola $x^2 = 4ay$. The tangents to the parabola at P and Q intersect at R . The normals at P and Q intersect at T . The point M is the midpoint of the chord PQ . The point S is the focus $(0, a)$.

- (i) Find the coordinates of M . 1

$$M \text{ is at } \left(\frac{2ap + 2aq}{2}, \frac{ap^2 + aq^2}{2} \right) = \left(a(p + q), \frac{1}{2}a(p^2 + q^2) \right)$$

- (ii) Show that $pq = -1$ if PQ is a focal chord. 2

$$\text{Equation of } PQ: y - ap^2 = \frac{p+q}{2}(x - 2ap)$$

Substitute the coordinates of S into the equation:

$$a - ap^2 = \frac{p+q}{2}(0 - 2ap)$$

$$a - ap^2 = -ap^2 - apq$$

$$a = -apq$$

$$pq = -1$$

- (iii) By considering the x -coordinates of the three points or otherwise, show that if PQ is a focal chord, then the points R , T and M are collinear.

3

THE LONG WAY:

If PQ is a focal chord, then the tangents meet on the directrix $y = -a$.

R lies on the tangent at P : $y = px - ap^2$. Let $y = -a$:

$$\begin{aligned} -a &= px - ap^2 \\ px &= ap^2 - a \\ &= a(p^2 - 1) \\ x &= \frac{a}{p}(p^2 - 1) \\ &= \frac{a}{p}(p^2 + pq) \\ &= a(p + q) \end{aligned}$$

So R has the same x -coordinate as M .

Find the equations of the normals:

$$\text{At } P, \quad x + py = ap^3 + 2ap$$

$$\text{Similarly, at } Q: \quad x + qy = aq^3 + 2aq$$

Solving these simultaneously to find T :

$$\begin{aligned} py - qy - ap^3 + aq^3 &= 2ap - 2aq \\ (p - q)y &= 2a(p - q) + a(p^3 - q^3) \\ &= 2a(p - q) + a(p - q)(p^2 + pq + q^2) \\ y &= 2a + a(p^2 + q^2 - 1) \\ x &= 2ap + ap^3 - py \\ &= 2ap + ap^3 - 2ap - ap^3 - apq^2 + ap \\ &= ap - apq^2 \\ &= ap + aq \\ &= a(p + q) \end{aligned}$$

So T has the same x -coordinate, and thus R , M and T all lie on the same vertical line.

THE SHORT WAY

$\angle QRP$, $\angle RPT$ and $\angle RQT$ are all right angles (tangents $\perp$ normals)

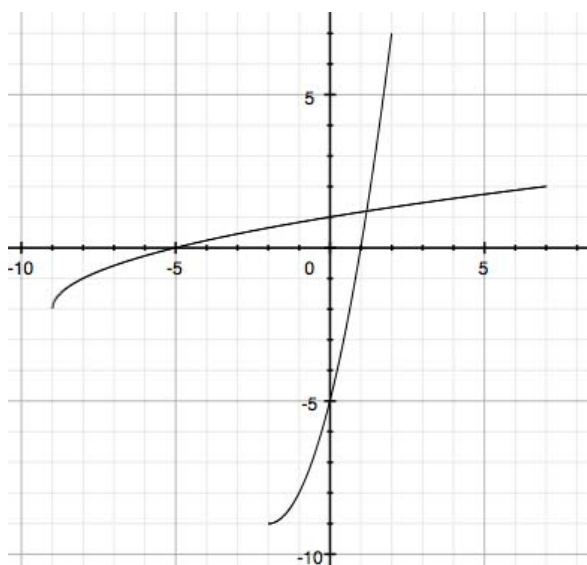
$\therefore$ The quadrilateral $RPTQ$ is a rectangle (all angles of quadrilateral are right angles)

$\therefore$ The midpoints of diagonals RT and PQ are the same (diagonals of a rectangle bisect each other)

$\therefore$ The points R , M and T must be collinear (midpoint M must lie on diagonal RT)

Question 13. (15 marks) (Start a new booklet)**Marks**(a) Consider the function $f(x) = (x+2)^2 - 9$, $-2 \leq x \leq 2$.(i) Find the equation of the inverse function $f^{-1}(x)$. **1**

$$\begin{aligned} \rightarrow x &= (y+2)^2 - 9 \\ x+9 &= (y+2)^2 \\ \sqrt{x+9} &= y+2 \\ y &= -2 + \sqrt{x+9} \end{aligned}$$

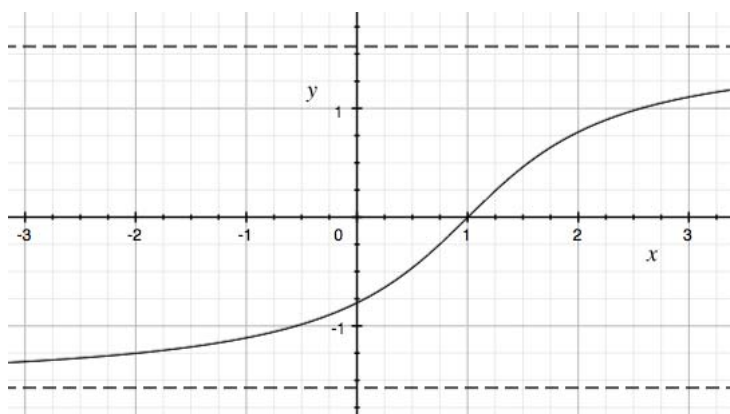
(ii) On the same diagram, sketch the graphs of $y = f(x)$ and $y = f^{-1}(x)$, showing clearly the coordinates of the endpoints and the intercepts on the coordinate axes. **3**(iii) Find the x -coordinate of the point of intersection of the curves $y = f(x)$ and $y = f^{-1}(x)$. **2**Find the intersection of $y = f(x)$ and $y = x$:

$$\begin{aligned} x &= (x+2)^2 - 9 \\ x &= x^2 + 4x + 4 - 9 \\ x^2 + 3x - 5 &= 0 \\ x &= \frac{-3 \pm \sqrt{3^2 - 4 \cdot 1 \cdot (-5)}}{2 \cdot 1} = \frac{\sqrt{29} - 3}{2} \end{aligned}$$

The lines intersect at $\left(\frac{\sqrt{29} - 3}{2}, \frac{\sqrt{29} - 3}{2} \right)$

(b) Consider the function $f(x) = \tan^{-1}(x-1)$.

- (i) Sketch the curve $y = f(x)$, showing clearly the equations of any asymptotes and the intercepts on the coordinate axes. 2



- (ii) Find the equation of the tangent to the curve $y = f(x)$ at the point where $x = 1$. 2

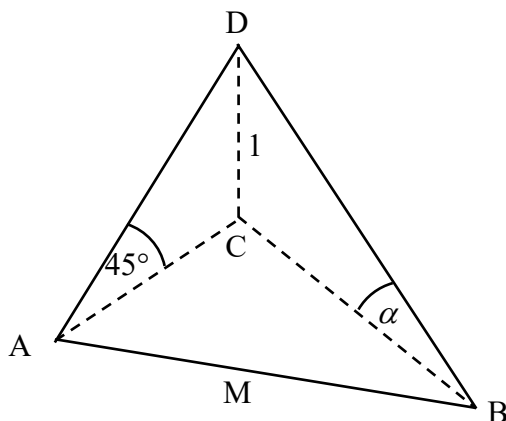
$$f(x) = \tan^{-1}(x-1)$$

$$f'(x) = \frac{1}{1+(x-1)^2} = \frac{1}{x^2 - 2x + 2}$$

$$f'(1) = \frac{1}{1^2 - 2(1) + 2} = 1$$

$$\text{Tangent: } y - 0 = 1(x - 1) \Rightarrow y = x - 1$$

(c)



CD is a vertical pole of height 1 metre that stands with its base C on horizontal ground. A is a point due South of C such that the angle of elevation of D from A is 45° . B is a point due East of C such that the angle of elevation of D from B is α . M is the midpoint of AB.

- (i) Show that $BC = \cot \alpha$ and hence show that $AB = \operatorname{cosec} \alpha$. 3

In $\triangle BCD$,

$$\frac{CD}{BC} = \tan \alpha$$

$$\frac{BC}{1} = \frac{1}{\tan \alpha}$$

$$BC = \cot \alpha$$

In $\triangle ABC$,

$$AB^2 = AC^2 + BC^2$$

$$= 1^2 + \cot^2 \alpha$$

$$= \operatorname{csc}^2 \alpha$$

$$AB = \operatorname{csc} \alpha$$

- (ii) Show that $CM = \frac{1}{2} \operatorname{cosec} \alpha$ 2

$\angle ACB = 90^\circ \therefore AB$ is the diameter of a circle with M as its centre
and A , B and C are equally distant from M

$$\therefore CM = AM = \frac{1}{2} AB = \frac{1}{2} \operatorname{csc} \alpha$$

Question 14. (15 marks) (Start a new booklet)

Marks

- (a) A particle is performing Simple Harmonic Motion in a straight line. At time t seconds, it has displacement x metres from a fixed point O in the line, velocity $v \text{ ms}^{-1}$ given by $v = -12 \sin\left(2t + \frac{\pi}{3}\right)$ and acceleration $\ddot{x} \text{ ms}^{-2}$. Initially the particle is 5 metres to the right of O .

- (i) Find an expression for x . 1

$$x = 6 \cos\left(2t + \frac{\pi}{3}\right) + c$$

When $t = 0$, $x = 5$

$$5 = 6 \cos \frac{\pi}{3} + c \Rightarrow c = 2$$

$$x = 6 \cos\left(2t + \frac{\pi}{3}\right) + 2$$

- (ii) Show that $\ddot{x} = -4(x - 2)$. 2

$$\ddot{x} = -24 \cos\left(2t + \frac{\pi}{3}\right)$$

$$= -4 \times 6 \cos\left(2t + \frac{\pi}{3}\right)$$

$$= -4(x - 2)$$

- (iii) Find the extremes of motion.

2

Let $v = 0$:

$$-12 \sin\left(2t + \frac{\pi}{3}\right) = 0$$

$$2t + \frac{\pi}{3} = 0, \pi, 2\pi, \dots$$

$$\begin{aligned} x &= 6 \cos\left(2t + \frac{\pi}{3}\right) + 2 \\ &= 6 \cos(0) + 2, 6 \cos(\pi) + 2 \\ &= 8, -4 \end{aligned}$$

- (iv) Find the time taken by the particle to return to its starting point for the first time.

2

Let $x = 5$:

$$6 \cos\left(2t + \frac{\pi}{3}\right) + 2 = 5$$

$$6 \cos\left(2t + \frac{\pi}{3}\right) = 3$$

$$\cos\left(2t + \frac{\pi}{3}\right) = \frac{1}{2}$$

$$2t + \frac{\pi}{3} = \frac{\pi}{3}, \frac{5\pi}{3}, \dots$$

$$2t = 0, \frac{4\pi}{3}$$

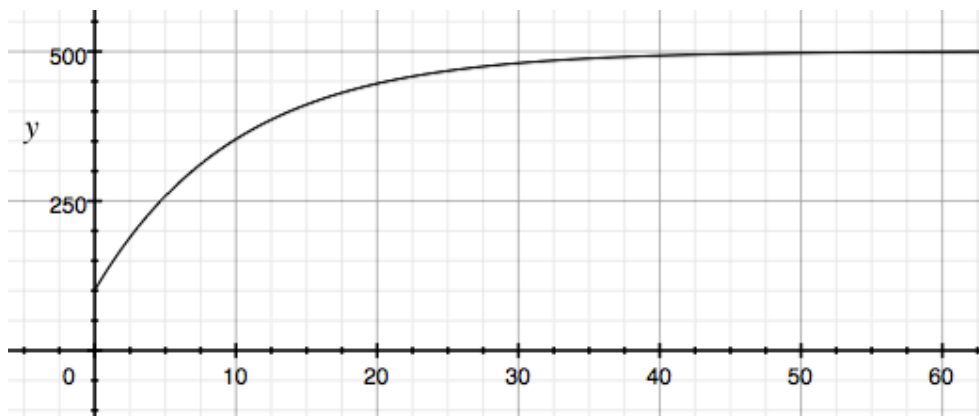
$$t = 0, \frac{2\pi}{3}$$

The particle takes $\frac{2\pi}{3}$ seconds to return to its starting point for the first time

- (b) After
- t
- hours, the number of individuals in a population is given by
- $N = 500 - 400e^{-0.1t}$
- .

- (i) Sketch the graph of
- N
- as a function of
- t
- , showing clearly the initial population size and the limiting population size.

2



- (ii) Show that $\frac{dN}{dt} = 0.1(500 - N)$. 1

$$\begin{aligned} N &= 500 - 400e^{-0.1t} \\ \frac{dN}{dt} &= -400e^{-0.1t} \times -0.1 \\ &= 0.1(400e^{-0.1t}) \\ &= 0.1(500 - N) \end{aligned}$$

- (iii) Find the population size for which the rate of growth of the population is half the initial rate of growth. 1

$$\text{Initial rate of growth } \frac{dN}{dt} = 0.1(500 - 100) = 40$$

$$\text{Let } \frac{dN}{dt} = 20 :$$

$$0.1(500 - P) = 20$$

$$500 - P = 200$$

$$P = 300$$

- (c) A particle is moving in a straight line. After time t seconds, it has displacement x metres from a fixed point O in the line, velocity $v \text{ ms}^{-1}$ given by $v = \sqrt{x}$, and acceleration $a \text{ ms}^{-2}$. Initially the particle is 1 metre to the right of O .

- (i) Show that a is constant. 1

$$\begin{aligned} \ddot{x} &= \frac{d}{dx} \left(\frac{1}{2} v^2 \right) \\ &= \frac{d}{dx} \left(\frac{1}{2} x \right) \\ &= \frac{1}{2} \end{aligned}$$

- (ii) Express x in terms of t . 2

$$\frac{dx}{dt} = x^{1/2}$$

$$\frac{dt}{dx} = \frac{1}{x^{1/2}} = x^{-1/2}$$

$$t = 2x^{1/2} + c$$

$$\text{When } t = 0, x = 1: \quad c = -2$$

$$t = 2x^{1/2} - 2$$

$$\frac{1}{2}(t + 2) = x^{1/2}$$

$$x = \frac{1}{4}(t + 2)^2$$

- (iii) Find the distance travelled by the particle during the third second of motion. 1

$$\text{When } t = 2, x = 4. \text{ When } t = 3, x = 6.25$$

The particle does not change direction in this time (v is always positive)

The particle travels 2.25 m during the third second.

END OF EXAM